Machine Learning
Quiz Refresher: Chapters 1–3, Questions

What it means to learn, from representations to decisions, from fitting to generalization; answers in the separate answer sheet

Machine Learning

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How to use this document

Three rounds, answers in the separate answer sheet. Round A is 30 one-line recall questions; if one takes more than a minute, the chapter section it comes from needs a reread. Round B is eight multiple-choice questions, letter plus a one-line justification. Round C is six exam-style problems with new numbers, so the chapters' worked examples will not carry you, the formulas must. Budget 25 minutes for A and B, 70 minutes for C. Work with the cheat sheet closed first, then open it to check.

Round A: rapid recall

Chapter 1.

A1. Define learning in one sentence. Name the two routes to giving a machine an ability.

A2. Why does having an ability not imply being able to write down the rules for it? Give one example.

A3. Name the three forms of experience and the learning paradigm each one leads to. What decides the paradigm?

A4. List the four ingredients of any learning system.

A5. What distinguishes regression from classification? Then distinguish prediction, decision and action.

A6. Feature engineering versus representation learning: what is the difference? What is self-supervision, and how does next-token prediction turn it into generation?

A7. What is a shortcut? Give one example and say why training performance does not reveal it.

A8. Write the entropy $H(p)$, the cross-entropy $H(p, q)$ and $\mathrm{KL}(p \,\|\, q)$. State Gibbs' inequality and what it implies about minimising cross-entropy.

Chapter 2.

A9. How does a categorical attribute become part of a vector $x \in \mathbb{R}^d$? In what sense does the representation bound what can be learned?

A10. Write the Bayes decision rule. Why does being optimal at each x separately make it optimal overall?

A11. Under squared loss, absolute loss and zero–one loss, what does the Bayes rule report?

A12. With costs C_{FP} and C_{FN}, when does the Bayes rule declare the positive class? Which way does the threshold move when misses are expensive?

A13. Write the kNN regression and classification rules. What does k control?

A14. Why must features be scaled before kNN? Write the Minkowski distance and name its $p = 2$ case.

A15. Write the curse-of-dimensionality formula for the probability of finding a neighbour within 0.1 on every coordinate. What does it say about locality in high dimension?

A16. Define a hypothesis space $\mathcal{H}$. What is the difference between a parameter and a hyperparameter?

A17. Write the population risk and the empirical risk. Which one can we compute, and which one do we want?

A18. Why is the training risk of the fitted model optimistic? Write the chain of inequalities that says so.

A19. Why do we not minimise the zero–one empirical risk directly for classification? What do we minimise instead?

A20. At a fixed x, the expected cross-entropy of a model $\hat{p}_\theta(x)$ equals what? What does this imply about the probabilities cross-entropy trains?

A21. Name the three subsets of the data and the job of each. How many times may the test set be used?

A22. Write the K-fold cross-validation estimate. What do you do once the hyperparameter is chosen?

Chapter 3.

A23. Define the generalization gap of a fitted model.

A24. What is inductive bias? Name the five channels through which it enters.

A25. For nested families $\mathcal{H}_1 \subset \mathcal{H}_2 \subset \cdots$, what happens to the best training error as capacity grows? And to validation error?

A26. Give the training/validation symptoms of underfitting and of overfitting.

A27. Define bias and variance at a point x and write the decomposition of the expected squared error.

A28. Write regularised ERM and its constrained equivalent. What does L_2 do to coefficients that L_1 does not, and vice versa?

A29. In early stopping, what is the hyperparameter and how is it chosen? What is patience for?

A30. Define leakage and give three forms. Which split is right for "events next month", and which for "a hospital never seen"? Why is accuracy misleading under imbalance, and what replaces it?

Round B: multiple choice, letter plus justification

Each question is 2 points: 1 for the letter, 1 for a one-line justification.

B1. **Cost threshold.** A false alarm costs 1, a miss costs 9. The Bayes rule declares the positive class when $p(x)$ exceeds:
(a) 0.1 (b) 0.111 (c) 0.5 (d) 0.9

B2. **Which number to report.** At some x, $Y = 0$ with probability 0.8 and $Y = 10$ with probability 0.2. The Bayes prediction under squared loss is:
(a) 0 (b) 2 (c) 5 (d) 10

B3. **Curse.** $d = 10$ features, $n = 10^6$ training points uniform in the unit cube. The probability that some point lies within 0.1 of a query on every coordinate is about:
(a) 1 (b) 0.63 (c) 10^{-4} (d) exactly 0

B4. **Extreme k.** kNN with $k = N$ predicts, for every query x:
(a) the label of the nearest point (b) the global majority class (or global mean), whatever x is (c) the Bayes-optimal class (d) nothing: the rule is undefined

B5. **Expected cross-entropy.** At x the true posterior is $p = (0.7, 0.3)$ and the model says $q = (0.5, 0.5)$. The expected cross-entropy $\mathbb{E}[-\log q_Y \mid x]$, in nats, is:
(a) 0.611 (b) 0.693 (c) 0.082 (d) 0.5

B6. **Not leakage.** Which of these is *not* a form of leakage?
(a) standardising with the mean and variance of all the data before splitting (b) the same patient in training and test (c) choosing λ by its validation error (d) a feature that is filled in after the outcome is known

B7. **Lasso in one dimension.** The unregularised estimate is $z = 0.8$ and $\lambda = 1$. The lasso solution $\arg\min_w \frac{1}{2}(w - z)^2 + \lambda|w|$ is:
(a) 0.8 (b) -0.2 (c) 0 (d) 0.4

B8. **Diagnosis.** Model A: training error 2%, validation 15%. Model B: training 14%, validation 15%. Which is right?
(a) choose A, its training error is lower (b) they are equivalent, same validation error (c) choose B, its gap is smaller (d) A is overfitting and B is closer to underfitting; regularise A or add capacity to B, then compare on validation again

Round C: exam-style problems

C1. Three actions, two costs [8 pts]

A hospital alarm system receives, for each patient, the posterior probability $p = \mathbb{P}(Y = 1 \mid x)$ that the patient is deteriorating. Three actions are available: *alarm* (cost $C_{\text{FP}} = 2$ if the patient is fine, 0 otherwise), *no alarm* (cost $C_{\text{FN}} = 18$ if the patient is deteriorating, 0 otherwise), and *order a test*, which costs 1.2 whatever the truth and then settles the case perfectly.

(a) With only alarm / no alarm available, at which p does the Bayes rule switch to alarm? Show the inequality.

(b) Compute the expected cost of each of the three actions for $p = 0.05$, $p = 0.12$ and $p = 0.5$, and give the Bayes action for each patient.

(c) For which range of p is ordering the test the Bayes action? Show the two inequalities.

(d) The system's designers trained the model with the cross-entropy loss. Why is that not a contradiction with the decision loss being costs of 2, 18 and 1.2?

C2. Which number to report [8 pts]

At a fixed input x, the output Y takes the value 1 with probability 0.4, the value 2 with probability 0.3 and the value 10 with probability 0.3.

(a) Give the Bayes prediction under squared loss, under absolute loss and under zero–one loss (treat the three values as classes). Three different numbers?

(b) Compute the expected squared loss of the squared-loss prediction and of the absolute-loss prediction. Check the identity $\mathbb{E}[(Y - a)^2 \mid x] = \text{Var}(Y \mid x) + (m(x) - a)^2$ on the second one.

(c) Compute the expected absolute loss of both predictions. Which is smaller?

(d) A client says "I would rather overshoot than undershoot: a shortfall costs me nine times an excess". Which quantile does the pinball loss ask for, and what is it here?

C3. Nearest neighbours and the scale of a feature [10 pts]

Five customers are described by (income in k$, age) with a label:

	$x^{(1)}$	$x^{(2)}$	$x^{(3)}$	$x^{(4)}$	$x^{(5)}$
income	20	25	80	85	90
age	60	58	25	30	62
label	A	A	B	B	A

A new customer has income 82 and age 60.

(a) Compute the five Euclidean distances on the raw features and give the 3NN prediction.

(b) Rescale: divide income differences by $s_1 = 30$ and age differences by $s_2 = 15$. Recompute the distances and the 3NN prediction. Explain what changed and why.

(c) On the scaled features, what do 1NN and 5NN predict? Which of the three values of k is the most "underfitted"?

(d) The company has 10^6 customers described by $d = 20$ standardised features. Estimate the probability that a query has a training point within 0.1 on every coordinate. What does this say about kNN here?

C4. Least squares, interpolation and ridge [10 pts]

Training data: $(x, y) \in \{(0,1), (1,3), (2,3), (3,5)\}$.

(a) Fit $y \approx wx + b$ by least squares with the closed-form formulas. Give the four residuals and the training mean squared error.

(b) A degree-3 polynomial can be fitted to these four points. What is its training error, and why is that not evidence it is a better model? Name the proposition behind the answer.

(c) Centre the data ($\bar{x} = 1.5$, $\bar{y} = 3$) and fit $y_c \approx wx_c$ without intercept. Check you recover w from (a). Then compute the ridge estimate $\hat{w}_\lambda = \sum x_c y_c / (\sum x_c^2 + N\lambda)$ for $\lambda = 0.25$ and $\lambda = 1$. What happens as $\lambda \to \infty$?

(d) A validation point $(4, 5.5)$ arrives. Compute the squared validation error of the least-squares line. Would you expect the interpolating cubic to do better or worse at $x = 4$, and why?

C5. Capacity, selection and the bias–variance budget [10 pts]

Polynomial regressors of increasing degree were fitted to the same training set and scored on a validation set:

degree	1	2	3	5	9
training MSE	0.30	0.18	0.12	0.06	0.00
validation MSE	0.32	0.20	0.15	0.19	0.41

(a) Which degrees underfit, which overfit, and which do you select? Justify with the two columns.

(b) Is 0.15 an unbiased estimate of the selected model's test error? Write the inequality that answers this, and say what the test set is for.

(c) You evaluate on the test set once and get 0.17. You are disappointed. May you go back and try degree 4? Why?

(d) At one input x the true value is $f^*(x) = 5$, the noise variance is 0.2. Over many training sets, the degree-9 model predicts on average 4.9 with variance 0.5; a ridge-regularised degree-9 model predicts on average 4.6 with variance 0.05. Compute the expected squared error of each at x and say which trade was worth it.

(e) Explain in two sentences why the ridge penalty reduced variance and increased bias here.

C6. Evaluating a fraud detector honestly [10 pts]

A test set has 1,000 card transactions, 50 of them fraudulent. A detector flags 80 transactions: 40 of the flags are real fraud and 40 are legitimate.

(a) Fill the confusion matrix (TP, FP, FN, TN).

(b) Compute accuracy, recall, precision, specificity, balanced accuracy and F_1.

(c) A "detector" that flags nothing: compute its accuracy and balanced accuracy. Which metric exposes it?

(d) The bank wants to deploy the model on next month's transactions. Which split should the evaluation have used, and what goes wrong with a uniformly random split?

(e) A feature `chargeback_filed` predicts fraud almost perfectly in the historical data. Why must it be removed?

(f) The team standardises features with the mean and variance of the whole dataset, then runs 5-fold cross-validation. Name the error and state the correct procedure.